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MULTILEVEL INVERTER AND ASSOCIATED METHOD

Abstract

Provided is a multilevel inverter for converting a DC voltage into an AC voltage or an AC voltage into a DC voltage, including at least one phase (P) and a method for controlling the same, the multilevel inverter includes a set of capacitors including N capacitors, N being an integer greater than or equal to three, the multilevel inverter having N+1 levels, a balancing device to balance the voltage on the terminals of each of the capacitors of the set of capacitors, and a switching module for switching the input DC voltage and being able to convert the input DC voltage into the respective phase of the output AC voltage.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS
[0001] This application claims priority to European Patent Application Serial Number EP23214190.3, filed Dec. 5, 2023, which is herein incorporated by reference.
TECHNICAL FIELD
[0002] The present invention concerns multilevel inverters and methods for controlling a multilevel inverter.
[0003] Generally, a multilevel inverter supplies a load from a DC voltage source or supplies a DC from an AC source.
BACKGROUND
[0004] The load may have more phases, each phase of the load being connected to an output terminal of the multilevel inverter.
[0005] FIG. **1** illustrates an example of a multilevel inverter supplying one phase of a load **1**.
[0006] The inverter comprises a set of capacitors **2** comprising three capacitors extending between two input terminals **3**, **4** and connected together in series through two intermediate points **5**, **6**.
[0007] The inverter further comprises a balancing device **7** known from the document US2013/0114320 balancing the voltage in each of the intermediate points **5**, **6** of the set of capacitors **2** regardless of the voltage and of the current delivered on the two input terminals **3**, **4**, and a switching module **8**.
[0008] The multilevel inverter delivers four voltage levels.
[0009] The balancing device **7** and the switching module **8** are connected to the input terminals **3**, **4** and both intermediate points **5**, **6**.
[0010] The switching module **8** delivers an AC voltage on an output terminal **9** of the inverter.
[0011] The voltage between the two input terminals **3**, **4** is equal to VDC.
[0012] A first input terminal **3** is used as a voltage reference. The voltage at a first intermediate point **5** is equal to VDC/2, the voltage at the second intermediate point **6** is equal to 2 VDC/3, and the voltage at the second intermediate point **6** is equal to VDC.
[0013] The switching module **8** comprises a first set S**1** of switches connecting the first input terminal **3** to the output terminal **9**, a second set S**2** of switches connecting the first intermediate point **5** to the output terminal **9**, a third set S**3** of switches connecting the second intermediate point **6** to the output terminal **9**, and a fourth set S**4** of switches connecting the second input terminal **4** to the output terminal **9**.
[0014] When the first set S**1** of switches is closed, the first input terminal **3** is connected to the output terminal **9**. The second, third and fourth set S**2**, S**3**, S**4** are open.
[0015] The voltage through the fourth set S**4** of open switches is equal to VDC, the voltage through the third set S**3** of open switches is equal to 2 VDC/3, and the voltage through the second set S**2** of open switches is equal to VDC/3.
[0016] When the fourth set S**4** of switches is closed, the second input terminal **4** is connected to the output terminal **9**. The first, second and fourth set S**1**, S**2**, S**3** are open.
[0017] The voltage through the first set S**1** of open switches is equal to VDC, the voltage through the second set S**2** of open switches is equal to 2 VDC/3, and the voltage through the third set S**3** of open switches is equal to VDC.
[0018] The first and fourth set S**1**, S**4** are designed to endure the voltage VDC, and the second and third sets S**2**, S**3** are designed to endure the voltage 2 VDC/3.
[0019] Furthermore, the second and third sets S**2**, S**3** have to withstand an inversion of the voltage

at their ends.
[0020] When the first set S**1** is closed, the end of the second and third sets S**2**, S**3** connected to the output terminal **9** has a potential lower than the potential of the end of the said sets connected to the intermediate points **5**, **6**.
[0021] When the fourth set S**4** is closed, the end of the second and third sets S**2**, S**3** connected to the output terminal **9** has a potential greater than the potential of the end of the said sets connected to the intermediate points **5**, **6**.
[0022] The first and fourth sets S**1**, S**4** comprise unidirectional switches E**1**, and the second and third sets S**2**, S**3** comprise bidirectional switches E**2** to withstand the voltage inversion.
[0023] Each bidirectional switch E**2** may comprise two unidirectional switches E**1** in a back-to-back configuration.
[0024] Each switch E**1** may endure a voltage VDC/3 so that the first and fourth set S**1**, S**4** comprise each three switches E**1** and the second and third sets S**2**, S**3** comprise each four switches E**1**.
[0025] The four-level inverter comprises fourteen switches.
[0026] For multilevel inverters delivering more than four levels, the number of switches E**1** increases dramatically so that the encumbrance of each switching module **8** increases and the controlling of the switches E**1** is more complex.
[0027] It is therefore proposed to remedy the disadvantage related to switching modules for multilevel inverters known from the prior art.
SUMMARY
[0028] In view of the foregoing the invention provided is a multilevel inverter for converting an input DC voltage into an output AC voltage or an AC voltage into a DC voltage, including at least one phase.
[0029] According to an embodiment, the multilevel inverter includes a positive input terminal, a negative input terminal, and an output terminal for the at least one phase, a set of capacitors including N capacitors connected in series between both input terminals and connected together through intermediate points, two extreme capacitors of the N capacitors being directly connected to one of the two input terminals, N being an integer so that N=Qn+1−1, Q being an integer equal or greater than two, and n being an integer equal or greater than one, the inverter having N+1 levels, a balancing device connected between both input terminals and the intermediate points, and configured to balance the voltage on the terminals of each of the capacitors of the set of capacitors regardless of the DC voltage and the current delivered on the positive and negative input terminals and regardless of the AC voltage and current delivered by the phase, for the output terminal, a module for switching the DC voltage and being able to convert the DC voltage on the positive and negative input terminals into the AC voltage on the output terminal, or for switching the AC voltage on the output terminal and being able to convert the AC voltage into the DC voltage on the positive and negative input terminals, each switching module comprising: input connections, each input connection being connected to one of the two input terminals or to one of the intermediate points, and supply lines, each supply lines connecting an input connection to the respective output terminal and comprising a plurality of switches connected together in series.
[0030] A set of switches of the plurality of switches of a first supply line connected to a first input connection is shared with a second supply line connected to a second input connection so that a first current flowing from the first input connection to the respective output terminal and a second current flowing from the second input connection to the respective output terminal flow through the set of switches of the plurality of switches.
[0031] Advantageously, the multilevel inverter further comprises a command circuit configured for switching the switches of each switching module to minimize the voltage at the ends of each switch when the said switch is open.
[0032] Advantageously, Q is equal to two, N+1 being further the number of input connections and the number of supply lines, each switching module comprising (N+1)*log(N+1)/log(2) switches,

each supply lines comprising N switches.
[0033] According to an embodiment, n is equal to 1 so that N is equal to three, the set of capacitors including a first, a second and a third capacitors, a first end of the first capacitor being connected to a first end of the second capacitor through a first intermediate point, a first end of the third capacitor being connected to the second capacitor through a second intermediate point, the first input connection being connected to the second end of the first capacitor, the second input connection being connected to the first intermediate point, a third input connection of a third supply line being connected to the second intermediate point, and a fourth input connection of a fourth supply line being connected to the second end of the third capacitor, the first supply line comprising a first, second and a third switches connected in series so that a first end of the first switch is connected to the first input connection, a second end of the first switch is connected to a first end of the second switch, a second end of the second switch is connected to a first end of the third switch, and a second end of the third switch is connected to the output terminal, the second supply line comprising a fourth switch, the second and the third switches connected in series, so that a first end of the fourth switch is connected to the first intermediate point, a second end of the fourth switch is connected to the first end of the second switch, the third supply line comprising a fifth switch, a sixth switch and a seventh switch connected in series, so that a first end of the fifth switch is connected to the second intermediate point, a second end of the fifth switch is connected to a first end of the sixth switch, a second end of the sixth switch is connected to a first end of the seventh switch, and a second end of the seventh switch is connected to the output terminal, and the fourth supply line comprising an eighth switch, the sixth and seventh switches connected in series so that a first end of the eighth switch is connected to the fourth input connection, a second end of the eighth switch is connected to the first end of the sixth switch.
[0034] Advantageously, the second end of the first capacitor is connected to the positive input terminal and the second end of the third capacitor is connected to the negative input terminal, the multilevel power inverter being a four-level inverter.
[0035] According to an embodiment, the multilevel inverter further comprises a plurality of balancing resistors, a first balancing resistor connecting the first and the second ends of the first switch, a second balancing resistor connecting the first and the second ends of the second switch, a third balancing resistor connecting the first and the second ends of the third switch, a fourth balancing resistor connecting the first and the second ends of the sixth switch, a fifth balancing resistor connecting the first and the second ends of the seventh switch, and a sixth balancing resistor connecting the first and the second ends of the eighth switch.
[0036] According to an embodiment, each switch comprises a unidirectional switch including at least a diode and at least a gate turn off thyristor, the anode of the gate turn off thyristor being connected to the cathode of the diode and the cathode of the a gate turn off thyristor being connected to the anode of the diode, the gate of the gate turn off thyristor being connected to the command circuit.
[0037] Advantageously, each switch comprises a unidirectional switch including at least a diode and at least a field effect transistor, wherein the drain of the transistor is connected to the cathode of the diode and the source of the transistor is connected to the anode of the diode, the gate of the transistor being connected to the command circuit.
[0038] According to an embodiment, Q is greater or equal than three, N+1 being further the number of input connections and the number of supply lines, some switches of the module comprising unidirectional switches and bidirectional switches.
[0039] Advantageously, the balancing device comprises two switching modules and an inductor, each input connection of the two switching modules of the balancing device being connected to one of the two input terminals or to one of the intermediate points, the output terminal of each switching module of the balancing device being connected to an end of the inductor.
[0040] Another embodiment of the invention relates to a method for controlling a multilevel

inverter converting an DC voltage into an AC voltage or an AC voltage into a DC voltage, including at least one phase.

[0041] The multilevel inverter includes a positive input terminal, a negative input terminal, and an output terminal for the at least one phase, a set of capacitors including N capacitors connected in series between both input terminals and connected together through intermediate points, two extreme capacitors of the N capacitors being directly connected to one of the two input terminals, N being an integer so that N=Qn+1−1, Q being an integer equal or greater than two, and n being an integer equal or greater than one, the inverter having N+1 levels, a balancing device connected between both input terminals and the intermediate points, and configured to balance the voltage on the terminals of each of the capacitors of the set of capacitors regardless of the DC voltage and the current delivered on the positive and negative input terminals and regardless of the AC voltage and current delivered by the phase, for the output terminal, a module for switching the DC voltage and being able to convert the DC voltage on the positive and negative input terminals into AC voltage on the output terminal, or for switching the AC voltage on the output terminal and being able to convert the AC voltage into the DC voltage on the positive and negative input terminals, each switching module includes input connections, each input connection being connected to one of the two input terminals, or to one of the intermediate points, and o supply lines, each supply lines connecting an input connection to the output terminal and comprising a plurality of switches connected together in series.

[0042] The method includes controlling a set of switches of the plurality of switches of a first supply line connected to a first input connection shared with a second supply line connected to a second input connection so that a first current flowing from the first input connection to the respective output terminal and a second current flowing from the second input connection to the respective output terminal flow through the set of switches of the plurality of switches.

[0043] Advantageously, Q is equal to two, N+1 being further the number of input connections and the number of supply lines, each switching module comprising log (N+1)/log(2)*(N+1) switches, each supply lines comprising N switches.

[0044] According to an embodiment, n is equal to 1 so that N is equal to three, the set of capacitors including a first, a second and a third capacitors, a first end of the first capacitor being connected to a first end of the second capacitor through a first intermediate point, a first end of the third capacitor being connected to the second capacitor through a second intermediate point, the first input connection being connected to the second end of the first capacitor, the second input connection being connected to the first intermediate point, a third input connection of a third supply line being connected to the second intermediate point, and a fourth input connection of a fourth supply line being connected to the second end of the third capacitor, the first supply line comprising a first, second and a third switches, the second supply line comprising a fourth switch, the second and the third switches, the third supply line comprising a fifth switch, a sixth switch and a seventh switch, and the fourth supply line comprising a eighth switch, the sixth and seventh switches, the method includes closing the fourth, sixth, seventh and eight switches and opening the first, second, third and fifth switches to deliver a first level on the output terminal, closing the fourth, fifth, sixth and seventh switches and opening the first, second, third and eighth switches to deliver a second level on the output terminal, closing the second, third, fourth and fifth switches, and opening the first, sixth, seventh and eighth switches to deliver a third level on the output terminal, and closing the first, second, third and fifth switches, and opening the fourth, sixth, seventh and eighth switches to deliver a fourth level on the output terminal.

[0045] Advantageously, the method further includes opening the eighth switch to switch from the first level to the second level, opening the sixth and seven switches to switch from the second level to the third level, and opening the first switch to switch from the third level to the fourth level.

[0046] Preferably, the second end of the first capacitor is connected to the positive input terminal and the second end of the third capacitor is connected to the negative input terminal, the multilevel power inverter being a four-level power inverter.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0047] Other characteristics and advantages of the invention will emerge on reading the following description of embodiments of the invention, provided solely by way of non-limiting examples and with reference to the drawings in which:

[0048] FIG. **1** illustrates schematically a multilevel inverter known from the prior art,

[0049] FIG. **2** illustrates schematically a first embodiment of a multilevel inverter according to the invention,

[0050] FIG. **3** illustrates schematically a second embodiment of the multilevel inverter according to the invention,

[0051] FIG. **4** illustrates schematically a first embodiment of unidirectional switches according to the invention,

[0052] FIG. **5** illustrates schematically a second embodiment of unidirectional switches according to the invention,

[0053] FIG. **6** illustrates schematically a third embodiment of the multilevel inverter according to the invention, and

[0054] FIG. **7** illustrates schematically a fourth embodiment of the multilevel inverter according to the invention.

[0055] FIG. **8** illustrates schematically a first embodiment of bidirectional switches according to the invention,

[0056] FIG. **9** illustrates schematically a second embodiment of bidirectional switches according to the invention, and

[0057] FIG. **10** illustrates schematically another embodiment of a balancing device according to the invention.

DETAILED DESCRIPTION

[0058] FIG. **2** illustrates schematically a first embodiment of a multilevel inverter **10**.

[0059] The multilevel inverter **10** is connected to a continuous DC voltage source **11** and is able to convert a DC voltage into an AC voltage and includes P phases connected to an electric machine **12**.

[0060] The multilevel inverter **10** is further able to convert an AC voltage into a DC voltage.

[0061] The multilevel inverter **10** is a voltage inverter with N+1 levels, wherein N is an integer.

[0062] The multilevel inverter **10** comprises a positive input terminal **13**, a negative input terminal **14**, and an output terminal **15** for each of the P phases. P is an integer greater than or equal to **1**. In the exemplary embodiment of FIG. **2**, P is equal to **3**.

[0063] The conversion system **10** comprises a set **16** of capacitors, a device **17** for balancing the voltage on the terminals of the capacitors of the set **16**, P modules **18** for switching the input DC voltage, and a command circuit **19** for controlling each switching module **18**.

[0064] The DC voltage source **11** comprises an input DC voltage VDC between the input terminals **13**, **14** of the inverter **10**.

[0065] The set **16** of capacitors includes N capacitors connected in series between both input terminals **13**, **14** and connected together through intermediate points.

[0066] The integer N is such that N=Qn+1−1, Q being an integer equal or greater than two, and n being an integer equal or greater than one.

[0067] The two extreme capacitors of the N capacitors are directly connected to one of the two input terminals **13**, **14**.

[0068] The electric machine **12** is, for example, a three-phase electric motor. The electric motor **12** is, for example, a synchronous motor. Alternatively, the motor **12** is an asynchronous motor.
[0069] The set of capacitors **26**, the balancing device **17** and the P switching modules **18** are electrically connected in parallel on each other between the input terminals **13**, **14**. Both input terminals **13**, **14** are common to the set **16** of capacitors, to the balancing device **17** and to the P switching modules **28**. Each switching module **28** is also connected to a respective output terminal **15**.
[0070] The balancing device **17** is further connected to each end of the N capacitors.
[0071] The balancing device **17** balances the voltage on the terminals of each of the capacitors of the set **16** of capacitors regardless of the input DC voltage and the current delivered on the positive and negative input terminals **13**, **14** and regardless of the AC voltage and current delivered by the phases P as explained in the document US 2013/0114320.
[0072] FIG. **3** illustrates schematically a second embodiment of the inverter **10** comprising a first embodiment of switching modules **18**.
[0073] For illustration only one switching module **18** is represented.
[0074] When Q is equal to two, the switching module **18** comprise N+1 input connections, N+1 number of supply lines, and (N+1)*log(N+1)/log(2) switches. Each supply line of the N+1 supply lines comprises (N+1)*log(N+1)/log(2) switches.
[0075] For illustration only, it is assumed that n is equal to one so that N is equal to three, the set **16** of capacitors including three capacitors C**1**, C**2**, C**3**.
[0076] A first end of the first capacitor C**1** is connected to a first end of the second capacitor C**2** through a first intermediate point P**1**.
[0077] A first end of the third capacitor C**3** is connected to the second capacitor C**2** through a second intermediate point P**2**.
[0078] The second end of the first capacitor C**1** is connected to the positive input terminal **13** and the second end of the third capacitor C**3** is connected to the negative input terminal **14**.
[0079] The multilevel power inverter **10** is a four-level inverter.
[0080] The balancing device **17** further comprises a first connection connected to the first intermediate point P**1** and a second connection connected to the second intermediate point P**2**. The switching module **18** comprises the N+**1** input connections **20**, **21**, **22**, **23** and the N+1 supply lines **24**, **25**, **26**, **27**.
[0081] The switching module **18** comprises the log (N+1)/log(2)*(N+1) switches **29**, **30**, **31**, **32**, **33**, **34**, **35**, **36**, each supply line **24**, **25**, **26**, **27** comprising N switches.
[0082] The switching module **18** further comprises an output connection **37** connected to the output terminal **15**.
[0083] A first supply line **24** connects a first input connection **20** connected to the second end of the first capacitor C**1** to the output connection **37** and comprises a first switch **29**, a second switch **30** and a third switch **31** connected together in series so that series so that a first end of the first switch **29** is connected to the first input connection **20**, a second end of the first switch **29** is connected to a first end of the second switch **30**, a second end of the second switch **30** is connected to a first end of the third switch **31**, and a second end of the third switch **31** is connected to the output connection **37**.
[0084] A second supply line **25** connects a second input connection **21** connected to the first intermediate point P**1** to the output connection **37** and comprises the second switch **30**, the third switch **31** and a fourth switch **32** connected together in series so that a first end of the fourth switch **32** is connected to the first intermediate point P**1**, a second end of the fourth switch **32** is connected to the first end of the second switch **30**.
[0085] A third supply line **26** connects a third input connection **22** connected to the second intermediate point P**2** to the output connection **37** and comprises a fifth switch **33**, a sixth switch **34** and a seventh switch **35** connected together in series so that a first end of the fifth switch **33** is

connected to the second intermediate point P**2**, a second end of the fifth switch **33** is connected to a first end of the sixth switch **34**, a second end of the sixth switch **34** is connected to a first end of the seventh switch **35**, and a second end of the seventh switch **35** is connected to the output connection **37**.

[0086] A fourth supply line **27** connects the fourth input connection **23** connected to the second end of the third capacitor C**3** to the output connection **37** and comprises the sixth switch **34**, the seventh switch **35** and an eighth switch **36** connected together in series so that so that a first end of the eighth switch **36** is connected to the fourth input connection **23**, a second end of the eighth switch **36** is connected to the first end of the sixth switch **34**.

[0087] The first embodiment of switching module **18** comprises a switching unit formed by the four supply lines **24**, **25**, **26**, **27**.

[0088] The balancing device **17** balances the voltage on the terminals of each of the capacitors C**1**, C**2**, C**3** of the set **16** so that the voltage at the first input connection **20** is equal to VDC, the voltage at the second input connection **21** is equal to 2 VDC/**3**, the voltage at the third input connection **22** is equal to VDC/3, and the voltage at the fourth input connection **23** is equal to nil.

[0089] The switches **29**, **30**, **31**, **32**, **33**, **34**, **35**, **36** may be identical and may be designed to endure at least a voltage equal to VDC/**3** between both ends of each switch when the said switch is open, each switch **29**, **30**, **31**, **32**, **33**, **34**, **35**, **36** may comprise a unidirectional switch.

[0090] The second switch **30** and the third switch **31** of the first supply line **24** are shared with the second supply line **25** so that a first current flowing from the first input connection **20** to the output terminal **15** and a second current flowing from the second input connection **21** to the output terminal **15** flow through the second switch **30** and the third switch **31**.

[0091] Similarly, the sixth switch **34** and the seventh switch **35** of the third supply line **26** are shared with the fourth supply line **27** so that a third current flowing from the third input connection **22** to the output terminal **15** and a fourth current flowing from the fourth input connection **23** to the output terminal **15** flow through the sixth switch **34** and the seventh switch **35**.

[0092] As the switches of the first and second sets of switches are shared respectively with the first and second supply lines **24**, **25**, and the third and fourth supply lines **26**, **27**, the number of switches of the switching module **18** is reduced compared to a switching module known from the prior art.

[0093] In this embodiment, the switching module **18** comprises eight switches whereas a switching module known from the prior art comprising four input connections comprises ten switches.

[0094] The switching module **18** may further comprise a plurality of balancing resistors R**1**, R**2**, R**3**, R**4**, R**5**, R**6**, the balancing resistors R**1**, R**2**, R**3**, R**4**, R**5**, R**6** having for example a same predetermined value.

[0095] The balancing resistors R**1**, R**2**, R**3**, R**4**, R**5**, R**6** permit to limit the voltage between the first and second ends of the switches **29**, **30**, **31**, **32**, **33**, **34**, **35**, **36** at a voltage equal to the voltage between both ends of a capacitor C**1**, C**2**, C**3** equal to VDC/3 when the said switches are opened.

[0096] A first balancing resistor R**1** connects the first and the second ends of the first switch **29**, a second balancing resistor R**2** connects the first and the second ends of the second switch **30**, a third balancing resistor R**3** connects the first and the second ends of the third switch **31**, a fourth balancing resistor R**4** connects the first and the second ends of the sixth switch **34**, a fifth balancing resistor R**5** connects the first and the second ends of the seventh switch **35**, and the sixth balancing resistor R**6** connects the first and the second ends of the eighth switch **36**.

[0097] The switches **29**, **30**, **31**, **32**, **33**, **34**, **35**, **36** may be unidirectional switches.

[0098] FIG. **4** illustrates schematically a first embodiment of a unidirectional switch.

[0099] The unidirectional switch includes a diode **40** and a field effect transistor **41**.

[0100] The field effect transistor **41** may be an insulated gate bipolar transistor IGBT or another type of field effect transistor.

[0101] The gate of the transistor **41** is connected to the command circuit **19**, the drain of the transistor **41** is connected to the cathode of the diode **40** and the source of the transistor **41** is connected to the anode of the diode **40**.

[0102] The transistor **41** is designed to support a voltage at least equal to VDC/3.

[0103] Each switch may comprise a plurality of transistors **41** in series so that the drain of one transistor is connected to the source of another transistor, the number of transistors **41** being determined so that the set of transistors **41** in series supports a voltage at least equal to VDC/3.

[0104] FIG. **5** illustrates schematically a second embodiment of the unidirectional switch.

[0105] The unidirectional switch includes the diode **40** and a gate turn off thyristor **42**.

[0106] The gate of the gate turn off thyristor **42** is connected to the command circuit **19**, the anode of the gate turn off thyristor **42** is connected to the cathode of the diode **40**, and the cathode of the gate turn off thyristor **42** is connected to the anode of the diode **40**.

[0107] The gate turn off thyristor **42** is designed to support a voltage at least equal to VDC/3.

[0108] Each switch may comprise a plurality of gate turn off thyristors **42** in series so that the anode of one thyristor is connected to the cathode of another thyristor, the number of thyristors **113** being determined so that the set of thyristors **42** in series supports a voltage at least equal to VDC/3.

[0109] The anode of the diode **40** of the first switch **29** is connected to the first input connection **20**, the anode of the diode **40** of the fourth switch **32** is connected to the second input connection **21**, the anode of the diode **40** of the fifth switch **33** is connected to the third input connection **22**, and the anode of the diode **40** of the eighth switch **36** is connected to the fourth input connection **23**.

[0110] The anode of the diode **40** of the second switch **30** and the anode of the diode **40** of the fourth switch **31** are respectively connected to the cathode of the diode **40** of the first switch **29** and the cathode of the diode **40** of the second switch **30**.

[0111] The anode of the diode **40** of the sixth switch **34** and the anode of the diode **40** of the seventh switch **31** are respectively connected to the cathode of the diode **40** of the fifth switch **33** and the cathode of the diode **40** of the sixth switch **34**.

[0112] The command circuit **19** controls the switches **29**, **30**, **31**, **32**, **33**, **34**, **35**, **36** of the switching module **18** and may control the switches **29**, **30**, **31**, **32**, **33**, **34**, **35**, **36** of the switching module **18** to minimize the voltage at the ends of each switch **29**, **30**, **31**, **32**, **33**, **34**, **35**, **36** when the said switch is open as described in the following.

[0113] It is assumed that the switches **29**, **30**, **31**, **32**, **33**, **34**, **35**, **36** of the switching module **18** are open so that no current flows through each switch **29**, **30**, **31**, **32**, **33**, **34**, **35**, **36**.

[0114] To deliver the first level on the output terminal **15**, the command circuit **19** closes the fourth, sixth, seventh and eight switches **32**, **34**, **35**, **36**, and opens the first, second, third and fifth switches **29**, **30**, **31**, **33**.

[0115] The first level is equal to nil.

[0116] As the fourth switch **32** is closed, the voltage between the output terminal **15** and a connection between the first, second and third switches **29**, **30**, **32** is equal to 2 VDC/3. The set of second and third switches **30**, **31** endure a voltage equal to 2 VDC/3 between the output terminal **15** and the first switch **29**.

[0117] To deliver the second level on the output terminal **15**, the command circuit **19** closes the fourth, fifth, sixth and seventh switches **32**, **33**, **35**, **36**, and opens the first, second, third and eighth switches **29**, **30**, **31**, **36**.

[0118] The second level is equal to VDC/3.

[0119] As the fourth switch **32** is closed, the voltage between the output terminal **15** and a connection between the first, second and third switches **29**, **30**, **32** is equal to VDC/3. The set of second and third switches **30**, **31** endure a voltage equal to VDC/3 between the output terminal **15** and the first switch **29**.

[0120] To deliver the third level on the output terminal **15**, the command circuit **19** closes the second, third, fourth and fifth switches **30**, **31**, **32**, **33**, and opens the first, sixth, seventh and eighth switches **29**, **34**, **35**, **36**.

[0121] The third level is equal to 2 VDC/3.
[0122] As the fifth switch **33** is closed, the voltage between the output terminal **15** and a connection between the fifth, sixth and eighth switches **33**, **34**, **36** is equal to VDC/3. The set of fifth and sixth switches **34**, **35** endure a voltage equal to VDC/3 between the output terminal **15** and the fifth switch **33**.
[0123] To deliver the fourth level on the output terminal **15**, the command circuit **19** closes the first, second, third and fifth switches **29**, **30**, **31**, **33**, and opens the fourth, sixth, seventh and eighth switches **33**, **34**, **35**, **36**.
[0124] The third level is equal to VDC.
[0125] As the fifth switch **33** is closed, the voltage between the output terminal **15** and a connection between the fifth, sixth and eighth switches **33**, **34**, **36** is equal to 2 VDC/3. The set of fifth and sixth switches **34**, **35** endure a voltage equal to 2 VDC/3 between the output terminal **15** and the fifth switch **33**.
[0126] Each switch of the switching module **18** endures between its ends a voltage equal to at most VDC/3 when the said switch is open in order to minimize the said voltage preserving the said switch from a defect.
[0127] To switch from the first level to the second level, the control command **19** opens the eighth switch.
[0128] To switch from the second level to the third level, the control command **19** opens the sixth and seven switches.
[0129] To switch from the third level to the fourth level, the control command **19** opens the first switch.
[0130] In order to maintain the voltage between both ends of each switch to at most VDC/3, the control circuit **19** controls the switches of the switching module **18** in ascending or descending order of the levels.
[0131] FIG. **6** illustrates schematically a third embodiment of the power inverter **10** comprising a second example of the switching module **18**.
[0132] It is assumed that Q is equal to two and n is equal to two, N being equal to seven. According to other embodiments, n may be greater than two.
[0133] The set **16** comprises seven capacitors C**10**, C**11**, C**12**, C**13**, C**14**, C**15**, C**16**.
[0134] The switching module **18** comprise eight input connections **45**, **46**, **47**, **48**, **49**, **50**, **51**, **52**, eight supply lines, twenty-two switches **57** to **72**, each supply line comprising seven switches.
[0135] A first end of a first capacitor C**10** is connected to a first end of a second capacitor C**11** through a first intermediate point P**10**.
[0136] A first end of a third capacitor C**12** is connected to the second end of the second capacitor C**11** through a second intermediate point P**11**.
[0137] A first end of a fourth capacitor C**13** is connected to the second end of the third capacitor C**12** through a third intermediate point P**12**.
[0138] A first end of a fifth capacitor C**14** is connected to the second end of the fourth capacitor C**13** through a fourth intermediate point P**13**.
[0139] A first end of a sixth capacitor C**15** is connected to the second end of the fifth capacitor C**14** through a fifth intermediate point P**14**.
[0140] A first end of the seventh capacitor C**16** is connected to the second end of the sixth capacitor C**15** through a sixth intermediate point P**15**.
[0141] The second end of the first capacitor C**10** is connected to the positive input terminal **13** and the second end of the seventh capacitor C**16** is connected to the negative input terminal **14**.
[0142] The multilevel power inverter **10** comprising the switching module **18** and the set **16** comprising the seven capacitors is an eight-level inverter.
[0143] The switching module **18** comprises a first and second switching unit **55**, **56** identical to the switching unit described above comprising the first switch **57**, **58**, the second switch **59**, **60**, the

third switch **61**, **62**, the fourth switch **63**, **64**, the fifth switch **65**, **66**, the sixth switch **67**, **68**, the seven switch **69**, **70** and the eighth switch **71**, **72** forming the supply lines.
[0144] The switching units **55**, **56** further comprise the output connection **73**, **74**.
[0145] The first supply line of the first switching unit **55** comprises the switches **57**, **59**, **61** and extends between a first input connection **45** connected to the second end of the first capacitor C**10** and the output connection **73**.
[0146] The second supply line of the first switching unit **55** comprises the switches **63**, **59**, **61** and extends between a second input connection **46** connected to the first intermediate point P**10**.
[0147] The third supply line of the first switching unit **55** comprises the switches **65**, **67**, **69** and extends between a third input connection **47** connected to the second intermediate point P**11** and the output connection **73**.
[0148] The fourth supply line of the first switching unit **55** comprises the switches **71**, **67**, **69** and extends between a fourth input connection **48** connected to the third intermediate point P**12** and the output connection **73**.
[0149] The first supply line of the second switching unit **56** comprises the switches **58**, **60**, **62** and extends between a fifth input connection **49** connected to the fourth intermediate point P**13** and the output connection **74**.
[0150] The second supply line of the second switching unit **56** comprises the switches **64**, **60**, **62** and extends between a sixth input connection **50** connected to the fifth intermediate point P**14** and the output connection **74**.
[0151] The third supply line of the second switching unit **56** comprises the switches **66**, **68**, **70** and extends between a seventh input connection **51** connected to the sixth intermediate point P**15** and the output connection **74**.
[0152] The fourth supply line of the second switching unit **56** comprises the switches **72**, **68**, **70** and extends between the eighth input connection **52** connected to the second end of the seventh capacitor C**16** and the output connection **74**.
[0153] The switching module **18** further comprises two sets **75**, **76** of four switches connected together in series.
[0154] A first set **75** connects the output connection **73** of the first switching unit **55** to the output terminal **15**, and the second set **76** connects the output connection **74** of the first switching unit **56** to the output terminal **15**.
[0155] The balancing device **17** (not represented on FIG. **6**) balances the voltage on the terminals of each of the capacitors C**11**, C**12**, C**13**, C**14**, C**15**, C**16** of the set **16** so that the voltage at the first input connection **45** is equal to VDC, the voltage at the second input connection **46** is equal to 6 VDC/7, the voltage at the third input connection **47** is equal to 5 VDC/7, the voltage at the fourth input connection **48** is equal to 4 VDC/7, the voltage at the fifth input connection **49** is equal to 3 VDC/7, the voltage at the sixth input connection **50** is equal to 2 VDC/7 the voltage at the seventh input connection **51** is equal to VDC/7, and the voltage at the eighth input connection **52** is equal to nil.
[0156] The switches **57**, **58**, **59**, **60**, **61**, **62**, **63**, **64**, **65**, **66**, **67**, **68**, **69**, **70**, **71**, **72** and the switches of the first and second sets **75**, **76** may be identical and may be designed to endure at least a voltage equal to VDC/7 at the ends of each switch when the said switch is open.
[0157] The set of capacitors may have more than seven capacitors so that the switching module **18** may have more than two switching units.
[0158] The switches of the switching module **18** are controlled by the command circuit **19** to minimize the voltage at the ends of each switch when the said switch is open.
[0159] According to an embodiment, the control of the switches of the second embodiment of the switching module **18** from the first embodiment of the power inverter illustrated on FIG. **3** to minimize the voltage at the ends of each switch when the said switch is open.
[0160] The first switching unit **55** and the second switching unit **56** may comprise the balancing

resistors R**1**, R**2**, R**3**, R**4**, R**5**, R**6**, R**7**, R**8** (not represented in FIG. **6**) as defined above. FIG. **7** illustrates schematically a fourth embodiment of the power inverter **10** comprising a third embodiment of the switching module **18**.

[0161] In this embodiment, the integer Q is equal to three and for clarity reason n is equal to **1**. N is equal to eight. According to an embodiment, Q may be greater than three. In another embodiment, n may be greater than one.

[0162] The set **16** comprises eight capacitors C**20**, C**21**, C**22**, C**23**, C**24**, C**25**, C**26**, C**27**.

[0163] The switching module **18** comprise nine input connections **80** to **88** and nine supply lines.

[0164] A first end of a first capacitor C**20** is connected to a first end of a second capacitor C**21** through a first intermediate point P**20**.

[0165] A first end of a third capacitor C**22** is connected to the second end of the second capacitor C**21** through a second intermediate point P**21**.

[0166] A first end of a fourth capacitor C**23** is connected to the second end of the third capacitor C**22** through a third intermediate point P**22**.

[0167] A first end of a fifth capacitor C**24** is connected to the second end of the fourth capacitor C**23** through a fourth intermediate point P**23**.

[0168] A first end of a sixth capacitor C**25** is connected to the second end of the fifth capacitor C**24** through a fifth intermediate point P**24**.

[0169] A first end of the seventh capacitor C**26** is connected to the second end of the sixth capacitor C**25** through a sixth intermediate point P**25**.

[0170] A first end of the eighth capacitor C**27** is connected to the second end of the seventh capacitor C**26** through a seven intermediate point P**26**.

[0171] The second end of the first capacitor C**20** is connected to the positive input terminal **13** and the second end of the eighth capacitor C**27** is connected to the negative input terminal **14**.

[0172] The multilevel power inverter **10** comprising the switching module **18** and the set **16** comprising the eight capacitors is a nine-level inverter.

[0173] The switching module **18** comprises a first input connection **80** connected to the second end of the first capacitor C**20**, a second input connection **81** connected to the first intermediate point P**20**, a third input connection **82** connected to the second intermediate point P**21**, a fourth input connection **83** connected to the third intermediate point P**22**, a fifth input connection **84** connected to the fourth intermediate point P**23**, a sixth input connection **85** connected to the fifth intermediate point P**24**, a seventh input connection **86** connected to the sixth intermediate point P**25**, an eighth input connection **87** connected to the seventh intermediate point P**26**, and a ninth input connection **88** connected to the second end of the eight capacitor C**27**.

[0174] The switching module **18** further comprises three switching units **90**, **91**, **92**.

[0175] Each switching unit **90**, **91**, **92** comprises a first input **93**, **94**, **95**, a second input **96**, **97**, **98**, a third input **99**, **100**, **101**, and an output **102**, **103**, **104**.

[0176] The first input **93** of the first switching unit **90** is connected to the first input connection **80**, the second input **96** of the first switching unit **90** is connected to the second input connection **81**, and the third input **99** of the first switching unit **90** is connected to the third input connection **82**.

[0177] The first input **94** of the second switching unit **91** is connected to the fourth input connection **83**, the second input **97** of the second switching unit **91** is connected to the fifth input connection **84**, and the third input **100** of the second switching unit **91** is connected to the sixth input connection **85**.

[0178] The first input **95** of the third switching unit **92** is connected to the seventh input connection **86**, the second input **98** of the third switching unit **92** is connected to the eighth input connection **87**, and the third input **101** of the third switching unit **92** is connected to the ninth input connection **88**.

[0179] Each switching unit **90**, **91**, **92** comprises a first set **105**, **106**, **107** of two switches connected together in series and extending between the first input **93**, **94**, **95** and the output **102**, **103**, **104**, a switch **108**, **109**, **110** connecting the second input **96**, **97**, **98** to the output **102**, **103**, **104**, and a second set **111**, **112**, **113** of two switches connected together in series and extending between the third input **99**, **100**, **101** and the output **102**, **103**, **104**.

[0180] The switching module **18** further comprises a third set **114** of switches comprising six switches connected together in series and connecting the output **102** of the first switching unit **90** to the output terminal **15**, a fourth set **115** of switches comprising three switches connected together in series and connecting the output **103** of the second switching unit **91** to the output terminal **15**, and a fifth set **116** of switches comprising six switches connected together in series and connecting the output **104** of the third switching unit **104** to the output terminal **15**.

[0181] The switching module **18** comprises a first supply line extending between the first input connection **80** and the output terminal **15**, and comprising the first set **105** of the first switching unit **90** and the third set **114**.

[0182] The switching module **18** comprises a second supply line extending between the second input connection **81** and the output terminal **15**, and comprising the switch **108** of the first switching unit **90** and the third set **114**.

[0183] The switching module **18** comprises a third supply line extending between the third input connection **82** and the output terminal **15**, and comprising the second set **111** of the first switching unit **90** and the third set **114**.

[0184] The switching module **18** comprises a fourth supply line extending between the fourth input connection **83** and the output terminal **15**, and comprising the first set **106** of the second switching unit **91** and the fourth set **115**.

[0185] The switching module **18** comprises a fifth supply line extending between the fifth input connection **84** and the output terminal **15**, and comprising the switch **109** of the second switching unit **91** and the fourth set **115**.

[0186] The switching module **18** comprises a sixth supply line extending between the sixth input connection **85** and the output terminal **15**, and comprising the second set **112** of the second switching unit **91** and the fourth set **115**.

[0187] The switching module **18** comprises a seventh supply line extending between the seventh input connection **86** and the output terminal **15**, and comprising the first set **107** of the third switching unit **92** and the fifth set **116**.

[0188] The switching module **18** comprises an eighth supply line extending between the eighth input connection **87** and the output terminal **15**, and comprising the switch **110** of the third switching unit **92** and the fifth set **116**.

[0189] The switching module **18** comprises a ninth supply line extending between the ninth input connection **88** and the output terminal **15**, and comprising the second set **113** of the third switching unit **92** and the fifth set **116**.

[0190] The third, fourth and fifth sets **114**, **115**, **116** of switches are shared by many supply lines reducing the number of switches of the switching module **18**.

[0191] The switches of the switching module **18** are controlled by the command circuit **19** as explained in the following to minimize the voltage between both ends of each switch when the said switch is open.

[0192] The balancing device **17** (not represented on FIG. **7**) balances the voltage on the terminals of each of the capacitors C**20**, C**21**, C**22**, C**23**, C**24**, C**25**, C**26**, C**27** of the set **16** so that the voltage at the first input connection **80** is equal to VDC, the voltage at the second input connection **81** is equal to 7 VDC/8, the voltage at the third input connection **82** is equal to 3 VDC/4, the voltage at the fourth input connection **83** is equal to 5 VDC/8, the voltage at the fifth input connection **84** is equal to VDC/2, the voltage at the sixth input connection **85** is equal to 3 VDC/8 the voltage at the seventh input connection **86** is equal to VDC/4, the voltage at the eighth input connection **87** is equal to VDC/8, and the voltage at the ninth input connection **88** is equal nil.

[0193] The switches **108**, **109**, **110** of the switching units **90**, **91**, **92**, the switches of the first and second sets **105**, **106**, **107**, **111**, **112**, **113** of the switching units **90**, **91**, **92**, and the switches of the third, fourth and fifth sets **114**, **115**, **116** may be designed to endure at least a voltage equal to VDC/**8** at the ends of each switch when the said switch is open.

[0194] For each switching unit **90**, **91**, **92**, the switches of the switching unit which are adjacent to switches of the said switching unit included in two supply lines comprise bidirectional switches and the other switches of the said switching unit comprise unidirectional switches for example as described on FIGS. **4** and **5**.

[0195] The switches of the first sets **105**, **106**, **107** and second sets **111**, **112**, **113** of switches of the switching units **90**, **91**, **92** each comprises a unidirectional switch as described on FIGS. **4** and **5**.

[0196] The switches **108**, **109**, **110** of the switching units **90**, **91**, **92**, each comprises a bidirectional switch.

[0197] The switches of the switching module **18** which are adjacent to switches of the said switching module included in two supply lines comprise bidirectional switches and the other switches of the said switching module **18** comprise unidirectional switches for example as described on FIGS. **4** and **5**.

[0198] The switches of the third set **114** and fifth set **116** of switches of the switching module **18** each comprises a unidirectional switch as described on FIGS. **4** and **5**.

[0199] The switches of the fourth set **115** of switches of the switching module **18** each comprises a bidirectional switch.

[0200] FIG. **8** illustrates schematically a first embodiment of a bidirectional switch.

[0201] The bidirectional switch comprises a first field effect transistor **120**, a second field effect transistor **121**, a first diode **122**, and a second diode **123**.

[0202] The field effect transistors **120**, **121** may be an insulated gate bipolar transistor IGBT or another type of field effect transistor.

[0203] The gate of the first and second field transistors **120**, **121** is connected to the command circuit **19**.

[0204] The source of the first field effect transistor **120** is connected to the source of the second field effect transistor **121**.

[0205] The drain of the first field transistor **120** is connected to the cathode of the first diode **122** and the source of the first transistor **120** is connected to the anode of the first diode **122**.

[0206] The drain of the second field transistor **121** is connected to the cathode of the second diode **123** and the source of the second transistor **121** is connected to the anode of the second diode **123**.

[0207] The transistors **120**, **121** are designed to support a voltage at least equal to VDC/3.

[0208] FIG. **9** illustrates schematically a second embodiment of a bidirectional switch.

[0209] The bidirectional switch comprises the first diode **122**, the second diode **123**, a first gate turn off thyristor **124**, and a second gate turn off thyristor **125**.

[0210] The gate of the first and second gate turn off thyristors **124**, **125** is connected to the command circuit **19**.

[0211] The cathode of the first gate turn off thyristor **124** is connected to the cathode of the second gate turn off thyristor **125**.

[0212] The cathode of the first diode **122** is connected to the anode of the first gate turn off thyristor **124** and the cathode of the second diode **123** is connected to the anode of the second gate turn off thyristor **125**.

[0213] The anode of the first diode **122** and the anode of the second diode **123** are connected to the cathode of the first and second gate turn off thyristors **124**, **125**.

[0214] The gate turn off thyristors **124**, **125** are designed to support a voltage at least equal to VDC/3.

[0215] Turning back to FIG. **7**, to deliver the first level on the output terminal **15**, the command circuit **19** closes the second set **111**, **112**, **113** of the first, second and third switching units **90**, **91**, **92**, closes the fifth set **116**, and opens the first set **105**, **106**, **107** of the switching units **90**, **91**, **92**,

the switch **108**, **109**, **110** of the switching units **90**, **91**, **92**, and the third and fourth sets **114**, **115**.
[0216] The first level is equal to nil.
[0217] To deliver the second level on the output terminal **15**, the command circuit **19** closes the second set **111**, **112** of the first and second switching units **90**, **91**, the switch **110** of the third switching unit **92**, and the fifth set **116**.
[0218] The command circuit **19** further opens the first set **105**, **106**, **107** of the switching units **90**, **91**, **92**, the switch **108**, **109** of the first and second switching units **90**, **91**, the second set **113** of the third switching unit **92**, the third and fourth sets **114**, **115**.
[0219] The second level is equal to VDC/8.
[0220] To deliver the third level on the output terminal **15**, the command circuit **19** closes the second set **111**, **112** of the first and second switching units **90**, **91**, the first set **107** of the third switching unit **92**, and the fifth set **116**.
[0221] The command circuit **19** further opens the first set **105**, **106** of the first and second switching units **90**, **91**, the switch **108**, **109**, **110** of the switching units **90**, **91**, **92**, the second set **113** of the third switching unit **92**, and the third and fourth sets **114**, **115**.
[0222] The third level is equal to VDC/4.
[0223] To deliver the fourth level on the output terminal **15**, the command circuit **19** closes the second set **111**, **112** of the first and second switching units **90**, **91**, the first set **107** of the third switching unit **92**, and the fourth set **115**.
[0224] The command circuit **19** further opens the first set **105**, **106** of the first and second switching units **90**, **91**, the switch **108**, **109**, **110** of the switching units **90**, **91**, **92**, the second set **113** of the third switching unit **92**, and the third and fifth sets **114**, **116**.
[0225] The fourth level is equal to 3 VDC/8.
[0226] To deliver the fifth level on the output terminal **15**, the command circuit **19** closes the second set **111** of the first second switching unit **90**, the switch **109** of the second switching unit **91**, the first set **107** of the third switching unit **92**, and the fourth set **115**.
[0227] The command circuit **19** further opens the first set **105**, **106** of the first and second switching units **90**, **91**, the switches **108**, **110** of the first and third switching units **90**, **92**, the second set **112**, **113** of the second and third switching units **91**, **92**, and the third and fifth sets **114**, **116**.
[0228] The fifth level is equal to VDC/2.
[0229] To deliver the sixth level on the output terminal **15**, the command circuit **19** closes the second set **111** of the first second switching unit **90**, the first set **106**, **107** of the second and third switching units **91**, **92** and the fourth set **115**.
[0230] The command circuit **19** further opens the first set **105** of the first switching unit **90**, the switches **108**, **109**, **110** of the switching units **90**, **91**, **92** and the second set **112**, **113** of the second and third switching units **91**, **92**, and the third and fifth sets **114**, **116**
[0231] The sixth level is equal to 5 VDC/8.
[0232] To deliver the seventh level on the output terminal **15**, the command circuit **19** closes the second set **111** of the first second switching unit **90**, the first set **106**, **107** of the second and third switching units **91**, **92** and the third set **114**.
[0233] The command circuit **19** further opens the first set **105** of the first switching unit **90**, the switches **108**, **109**, **110** of the switching units **90**, **91**, **92** and the second set **112**, **113** of the second and third switching units **91**, **92**, and the fourth and fifth sets **115**, **116**.
[0234] The seventh level is equal to 3 VDC/4.
[0235] To deliver the eighth level on the output terminal **15**, the command circuit **19** closes the switch **108** of the first second switching unit **90**, the first set **106**, **107** of the second and third switching units **91**, **92** and the third set **114**.
[0236] The command circuit **19** further opens the first set **105** of the first switching unit **90**, the switches **109**, **110** of the second and third switching units **91**, **92** and the second set **112**, **113** of the

second and third switching units **91**, **92**, and the fourth and fifth sets **115**, **116**.
[0237] The seventh level is equal to 7 VDC/8.
[0238] To deliver the ninth level on the output terminal **15**, the command circuit **19** closes the first sets **105**, **106**, **107** of the switching units **90**, **91**, **92**, and the third set **114**.
[0239] The command circuit **19** further opens the switches **108**, **109**, **110** of the switching units **90**, **91**, **92**, the second set **111**, **112**, **113** of the switching units **90**, **91**, **92**, and the fourth and fifth sets **115**, **116**.
[0240] The ninth level is equal to VDC.
[0241] The voltage between the output terminal **15** and the output **102** of the first switching unit **90** is at most equal to 3 VDC/4 (first level), the voltage between the output terminal **15** and the output **103** of the second switching unit **91** is at most equal to 3 VDC/8 (first and ninth levels), and the voltage between the output terminal **15** and the output **104** of the third switching unit **92** is at most equal to 3 VDC/4 (ninth level).
[0242] The third set **114** and the fifth set **116** endure a voltage equal to at most 3 VDC/4, and the fourth set **115** endures a voltage equal to at most 3 VDC/8.
[0243] To switch from the first level to the second level, the control command **19** opens the second set **113** of the third switching unit **92**.
[0244] To switch from the second level to the third level, the control command **19** opens the switch **110** of the third switching unit **92**.
[0245] To switch from the third level to the fourth level, the control command **19** opens the fifth set **116**.
[0246] To switch from the fourth level to the fifth level, the control command **19** opens the second set **112** of the second switching unit **91**.
[0247] To switch from the fifth level to the sixth level, the control command **19** opens the switch **109** of the second switching unit **91**.
[0248] To switch from the sixth level to the seventh level, the control command **19** opens the fourth set **115**.
[0249] To switch from the seventh level to the eighth level, the control command **19** opens the second set **111** of the first switching unit **90**.
[0250] To switch from the eighth level to the ninth level, the control command **19** opens the switch **108** of the first switching unit **90**.
[0251] In order to maintain the voltage between both ends of each switch of the third, fourth and fifth sets **114**, **115**, **116**, the control circuit **19** controls the switches of the switching module **18** in ascending or descending order of the levels.
[0252] The balancing device **17** may comprises two switching modules **18** and an inductor.
[0253] Each input connection of the two switching modules **18** of the balancing device is connected to one of the two input terminals or to one of the intermediate points.
[0254] The output terminal **15** of each switching module **18** of the balancing device is connected to an end of the inductor.
[0255] FIG. **10** illustrates schematically another embodiment of the balancing device **17** comprises a first switching module **181**, a second switching module **182**, and an inductor **130**.
[0256] For clarity reason, the example of the balancing device **17** is designed for the second embodiment of the multilevel inverter illustrated on FIG. **3**.
[0257] The first switching module **181** and the second switching module **182** have the same architecture as the switching module **18** described in FIG. **3**.
[0258] The first switching module **181** comprises the first input connection **201**, the second input connection **211**, the third input connection **221**, the fourth input connection **231**, and the output terminal **151** connected to a first end of the inductor **130**.
[0259] The second switching module **182** comprises the first input connection **202**, the second input connection **212**, the third input connection **222**, the fourth input connection **232**, and the

output terminal **152** connected to a second end of the inductor **130**.
[0260] The first input connections **201**, **202** of the first and second switching modules **181**, **182** are connected to the second end of the first capacitor C**1**.
[0261] The second input connections **211**, **212** of the first and second switching modules **181**, **182** are connected to the first intermediate point P**1**.
[0262] The third input connections **221**, **222** of the first and second switching modules **181**, **182** are connected to the second end of the second intermediate point P**2**.
[0263] The fourth input connections **201**, **202** of the first and second switching modules **181**, **182** are connected to the second end of the third capacitor C**3**.

Claims

1. A multilevel inverter for converting a direct current (DC) voltage into an alternate current (AC) voltage or an AC voltage into a DC voltage, including at least one phase, comprising: a positive input terminal, a negative input terminal (**14**), and an output terminal for the at least one phase; a set of capacitors including N capacitors connected in series between both input terminals and connected together through intermediate points, two extreme capacitors of the N capacitors being directly connected to one of the two input terminals, N being an integer so that N=Qn+1−1, Q being an integer equal or greater than two, and n being an integer equal or greater than one; the inverter having N+1 levels, a balancing device connected between both input terminals and the intermediate points, and configured to balance the voltage on the terminals of each of the capacitors of the set of capacitors regardless of the DC voltage and the current delivered on the positive and negative input terminals and regardless of the AC voltage and current delivered by the phase, for the output terminal, a switching module for switching the DC voltage and being able to convert the DC voltage on the positive and negative input terminals into the AC voltage on the output terminal, or for switching the AC voltage on the output terminal and being able to convert the AC voltage into the DC voltage on the positive and negative input terminals, each switching module comprising: input connections, each input connection being connected to one of the two input terminals or to one of the intermediate points, and supply lines, each supply lines connecting an input connection to the output terminal and comprising a plurality of switches connected together in series, characterized in that a set of switches of the plurality of switches of a first supply line connected to a first input connection is shared at least with a second supply line connected to a second input connection so that a first current flowing from the first input connection to the respective output terminal and a second current flowing from the second input connection to the respective output terminal flow through the set of switches of the plurality of switches.
2. The multilevel inverter according to claim 1, further comprising a command circuit configured for switching the switches of each switching module to minimize the voltage at the ends of each switch when the said switch is open.
3. The multilevel inverter according to claim 2, wherein Q is equal to two, N+1 being further the number of input connections and the number of supply lines, each switching module comprising (N+1)*log(N+1)/log(2) switches, each supply lines comprising N switches.
4. The multilevel inverter according to claim 3, wherein n is equal to 1 so that N is equal to three, the set of capacitors including a first capacitor, a second capacitor and a third capacitor, a first end of the first capacitor being connected to a first end of the second capacitor through a first intermediate point, a first end of the third capacitor being connected to the second capacitor through a second intermediate point, the first input connection being connected to the second end of the first capacitor, the second input connection being connected to the first intermediate point, a third input connection of a third supply line being connected to the second intermediate point, and a fourth input connection of a fourth supply line being connected to the second end of the third capacitor, the first supply line comprising a first switch, a second switch and a third switch

connected in series so that a first end of the first switch is connected to the first input connection, a second end of the first switch is connected to a first end of the second switch, a second end of the second switch is connected to a first end of the third switch, and a second end of the third switch is connected to the output terminal, the second supply line comprising a fourth switch, the second switch and the third switch connected in series, so that a first end of the fourth switch is connected to the first intermediate point, a second end of the fourth switch is connected to the first end of the second switch, the third supply line comprising a fifth switch, a sixth switch and a seventh switch connected in series, so that a first end of the fifth switch is connected to the second intermediate point, a second end of the fifth switch is connected to a first end of the sixth switch, a second end of the sixth switch is connected to a first end of the seventh switch, and a second end of the seventh switch is connected to the output terminal, and the fourth supply line comprising an eighth switch, the sixth switch and the seventh switch connected in series so that a first end of the eighth switch is connected to the fourth input connection, a second end of the eighth switch is connected to the first end of the sixth switch.

5. The multilevel inverter according to claim 4, wherein the second end of the first capacitor is connected to the positive input terminal and the second end of the third capacitor is connected to the negative input terminal, the multilevel power inverter being a four-level inverter.

6. The multilevel inverter according to claim 5, further comprising a plurality of balancing resistors, a first balancing resistor connecting the first and the second ends of the first switch, a second balancing resistor connecting the first and the second ends of the second switch, a third balancing resistor connecting the first and the second ends of the third switch, a fourth balancing resistor connecting the first and the second ends of the sixth switch, a fifth balancing resistor connecting the first and the second ends of the seventh switch, and a sixth balancing resistor connecting the first and the second ends of the eighth switch.

7. The multilevel inverter according to claim 6, wherein each switch comprises a unidirectional switch including at least a diode and at least a gate turn off thyristor, the anode of the gate turn off thyristor being connected to the cathode of the diode and the cathode of the a gate turn off thyristor being connected to the anode of the diode, the gate of the gate turn off thyristor being connected to the command circuit.

8. The multilevel inverter according to claim 6, wherein each switch comprises an unidirectional switch including at least a diode and at least a field effect transistor, wherein the drain of the transistor is connected to the cathode of the diode and the source of the transistor is connected to the anode of the diode, the gate of the transistor being connected to the command circuit.

9. The multilevel inverter according to claim 2, wherein Q is greater or equal than three, N+1 being further the number of input connections and the number of supply lines, switches of the switching module comprising unidirectional switches and bidirectional switches.

10. The multilevel inverter according to claim 9, wherein the balancing device comprises two switching modules and an inductor, each input connection of the two switching modules of the balancing device being connected to one of the two input terminals or to one of the intermediate points, the output terminal of each switching module of the balancing device being connected to an end of the inductor.

11. A method for controlling a multilevel inverter converting a direct current (DC) voltage into an alternate current (AC) voltage or an AC voltage into a DC voltage including at least one phase, the multilevel inverter comprising: a positive input terminal, a negative input terminal, and an output terminal for the at least one phase; a set of capacitors including N capacitors connected in series between both input terminals and connected together through intermediate points, two extreme capacitors of the N capacitors being directly connected to one of the two input terminals, N being an integer so that $N=Qn+1-1$, Q being an integer equal or greater than two, and n being an integer equal or greater than or to 1; the inverter having N+1 levels, a balancing device connected between both input terminals and the intermediate points, and configured to balance the voltage on the

terminals of each of the capacitors of the set of capacitors regardless of the DC voltage and the current delivered on the positive and negative input terminals and regardless of the AC voltage and current delivered by the phase, for the output terminal, a switching module for switching the DC voltage and being able to convert the DC voltage on the positive and negative input terminals into AC voltage on the output terminal, or for switching the AC voltage on the output terminal and being able to convert the AC voltage into the DC voltage on the positive and negative input terminals, each switching module comprising: input connections, each input connection being connected to one of the two input terminals, or to one of the intermediate points, and supply lines, each supply lines connecting an input connection to the output terminal and comprising a plurality of switches connected together in series, the method comprises: controlling a set of switches of the plurality of switches of a first supply line connected to a first input connection shared with a second supply line connected to a second input connection so that a first current flowing from the first input connection to the respective output terminal and a second current flowing from the second input connection to the respective output terminal flow through the set of switches of the plurality of switches.
12. The method according to claim 11, wherein Q is equal to two, N+1 being further the number of input connections and the number of supply lines, each switching module comprising log (N+1)/log(2)*(N+1) switches, each supply lines comprising N switches.
13. The method according to claim 12, wherein n is equal to one so that N is equal to three, the set of capacitors including a first capacitor, a second capacitor and a third capacitor, a first end of the first capacitor being connected to a first end of the second capacitor through a first intermediate point, a first end of the third capacitor being connected to the second capacitor through a second intermediate point, the first input connection being connected to the second end of the first capacitor, the second input connection being connected to the first intermediate point, a third input connection of a third supply line being connected to the second intermediate point, and a fourth input connection of a fourth supply line being connected to the second end of the third capacitor, the first supply line comprising a first switch, a second switch and a third switch, the second supply line comprising a fourth switch, the second switch and the third switch, the third supply line comprising a fifth switch, a sixth switch and a seventh switch, and the fourth supply line comprising a eighth switch, the sixth and seventh switches, the method comprising: closing the fourth switch, the sixth switch, the seventh switch and the eight switch and opening the first switch, the second switch, the third switch and the fifth switch to deliver a first level on the output terminal, closing the fourth switch, the fifth switch, the sixth switch and the seventh switch and opening the first switch, the second switch, the third switch and the eighth switch to deliver a second level on the output terminal, closing the second switch, the third switch, the fourth switch and the fifth switch, and opening the first switch, the sixth switch, the seventh switch and the eighth switch to deliver a third level on the output terminal, and closing the first switch, the second switch, the third switch and the fifth switch, and opening the fourth switch, the sixth switch, the seventh switch and the eighth switch to deliver a fourth level on the output terminal.
14. The method according to claim 13, further comprising: opening the eighth switch to switch from the first level to the second level, opening the sixth switch and the seven switch to switch from the second level to the third level, and opening the first switch to switch from the third level to the fourth level.
15. The method according to claim 14, wherein the second end of the first capacitor is connected to the positive input terminal and the second end of the third capacitor is connected to the negative input terminal, the multilevel power inverter being a four-level power inverter.